ASGHINLABS ACADEMY

Mathematics

FULL LESSON NOTES

COMPLETE · LESSONS 1-218

LOWER SIXTH

GCE A Level · MINESEC Progression

20 chapters · every lesson of the official MINESEC progression sheet

LEARN · PRACTISE · SUCCEED

By **Mr Ayemelong S. Ghislain**

www.asghinlabs.com

Academic Year **2026-2027**

ASGHINLABS

L6

ASGHINLABS ACADEMY

Lesson notes · Workbooks · Interactive lessons · AI Tutor

AsghinLabs Academy helps Cameroonian students prepare for the GCE with study materials that follow the official MINESEC progression. These notes are one part of the collection. Everything below is available from our website.

Workbooks available

Subject	What is available
Mathematics	Form 1, Form 2 and Form 3 workbooks · Complete O Level pack · Complete A Level pack (Lower & Upper Sixth)
Computer Science	Form 1, Form 2 and Form 3 workbooks · Complete O Level pack · Complete A Level pack (Lower & Upper Sixth)
ICT	Complete A Level pack (ICT 0796) with GCE past questions

How to order: pick your subject and class on the website, tap "Add to order", then send the order on WhatsApp. No account is needed and no payment is taken on the website. We reply to confirm the total, payment and delivery. Schools and classes ordering 10 copies or more can ask for a group price.

Order online: www.asghinlabs.com/academy.html#workbooks

Order on WhatsApp: +237 682 402 876

ASGHINLABS AI TUTOR

Stuck on an example or a practice question in these notes? Ask the AsghinLabs AI Tutor to explain it step by step, at any time of day.

Open the AI Tutor: www.asghinlabs.com/tutor.html

More from AsghinLabs Academy

- **Full lesson notes:** Lower and Upper Sixth Mathematics, Pure Mathematics with Mechanics, Biology, Physics and Chemistry, aligned with the MINESEC progression
- **Free interactive lessons:** www.asghinlabs.com/lessons.html
- **Free notes and resources:** www.asghinlabs.com/academy.html#resources
- **Private classes and GCE exam preparation:** www.asghinlabs.com/academy.html
- **WhatsApp channel (updates and new materials):** Join the AsghinLabs channel

www.asghinlabs.com · WhatsApp +237 682 402 876 · asghinlabs@gmail.com

Table of Contents

How to use these notes 13

Chapter 1: Theory of Quadratic Functions and Quadratic Equations 14

Lesson 1: Quadratic expressions and equations 14
Definition and standard form · identifying a, b and c · roots of an equation · forming equations from problems

Lesson 2: Method of completing the square 16
Perfect squares · the four-step method · surd answers · the form $a(x + p)^2 + q$ and minimum/maximum values

Lesson 3: Quadratic formula method 18
Derivation from completing the square · using the formula · exact and decimal answers · the discriminant and no real roots

Lesson 4: Solving by factorisation 20
The zero-product rule · splitting the middle term · difference of two squares and common factors · word problems

Lesson 5: Relation between roots and coefficients of a quadratic equation 22
Sum of roots $-b/a$ · product of roots c/a · forming an equation from its roots · finding unknown coefficients

Lesson 6: Symmetric properties of quadratic equations 23
Symmetric expressions in α and β · the standard identities · equations with related roots (α^2, $1/\alpha$, $\alpha + k$, 2α)

Lesson 7: Properties of quadratic functions 25
The parabola · shape from the sign of a · vertex and axis of symmetry · intercepts · domain, range and increasing/decreasing intervals

Lesson 8: Graphs of quadratic functions 27
Five-step sketching method · sketching ∪ and ∩ curves · tables of values and accurate graphs · solving equations graphically

Lesson 9: Simple transformation of the form $f(x) = ax^2$, $a \in \mathbb{R}$ 29
$y = ax^2$ as a stretch parallel to the y-axis · narrower and wider graphs · reflection in the x-axis when $a < 0$ · images of points

Lesson 10: Simple transformation of the form $y = f(x \pm a)$, $a \in \mathbb{R}$ 31
Horizontal translations · the "opposite sign" rule · the vertex after a translation · finding $f(x + a)$ algebraically · effect on roots

Lesson 11: Simple transformation of the form $y = f(x) \pm a$ 33
Vertical translations $y = f(x) \pm a$ · combining stretches and translations · vertex form $a(x - h)^2 + k$

Lesson 12: Application of the nature of roots 34
The discriminant $\Delta = b^2 - 4ac$ · equal, distinct and no real roots · tangents to curves · expressions positive for all x

Lesson 13: Absolute value equations of the form $|f(x)| = a$ 35
Meaning of $|x|$ · solving $|f(x)| = a$ for $a > 0$, $a = 0$, $a < 0$ · equations with x on both sides and checking answers

Review Exercise: Theory of Quadratic Functions and Quadratic Equations 36
Mixed exam-style questions with answers covering the whole topic

Chapter 2: Equations and Inequations (Inequalities) 37

Lesson 14: Basic rules of solving inequalities 37
The four rules for operating on inequalities · reversing the sign · interval notation

Lesson 15: Linear inequalities 39
Solving linear inequalities · double inequalities · inequalities with fractions · integer solutions

Lesson 16: Quadratic inequalities 40
Critical values · using the graph of a quadratic to solve inequalities · "between" and "outside" solutions

Lesson 17: Solutions to quadratic inequalities (sign tables and harder cases) 41
Sign tables · inequalities with a negative x^2 coefficient · rational inequalities · expressions positive for all x

Lesson 18: Absolute value inequalities of the form $|f(x)| < a$ 42
$|f(x)| < a$ means $-a < f(x) < a$ · linear and quadratic examples · distance interpretation

Lesson 19: Absolute value inequalities of the form $|f(x)| > a$ 43
$|f(x)| > a$ means $f(x) < -a$ or $f(x) > a$ · solving and writing the union of intervals

Lesson 20: Absolute value inequalities of the forms $|f(x)| < |g(x)|$ and $|f(x)| = \pm f(x)$ 44
$|f(x)| < |g(x)|$ by squaring · $|f(x)| = f(x)$ and $|f(x)| = -f(x)$

Lesson 21: Solutions to inequalities involving two unknowns....45
Graphing linear inequalities in two unknowns · shading regions · vertices of a feasible region · maximising a linear expression

Review Exercise: Equations and Inequations (Inequalities)....46
Mixed exam-style questions with answers covering the whole topic

Chapter 3: Graphs of Absolute Value Functions....47

Lesson 22: Graphs of modulus (absolute value) functions 1: y = |f(x)|....47
Definition of |f(x)| · sketching y = |f(x)| by reflecting negative parts · using graphs to count solutions

Lesson 23: Graphs of modulus functions 2: y = f(|x|) and solving with graphs....49
Sketching y = f(|x|) · symmetry about the y-axis · solving modulus equations and inequalities graphically

Review Exercise: Graphs of Absolute Value Functions....50
Mixed exam-style questions with answers covering the whole topic

Chapter 4: Polynomials....51

Lesson 24: Operations on polynomials: addition....51
Degree, coefficients and terms of a polynomial · adding polynomials by collecting like terms

Lesson 25: Subtraction of polynomials....52
Subtracting polynomials · changing the signs of every term · degree of a difference

Lesson 26: Multiplication of polynomials....53
Multiplying polynomials term by term · finding a single coefficient without full expansion · degree of a product

Lesson 27: Division of polynomials....54
Long division of polynomials · synthetic division · quotient and remainder · P(x) = D(x)Q(x) + R(x)

Lesson 28: The factor theorem....55
Statement of the factor theorem · testing linear factors (x − a) and (ax − b)

Lesson 29: The remainder theorem....56
Statement of the remainder theorem · remainders on division by (x − a) and (ax − b)

Lesson 30: Application of the factor theorem....57
Finding unknown coefficients from given factors · solving cubic equations

Lesson 31: Application of the remainder theorem....58
Finding unknown coefficients from given remainders · remainder on division by a quadratic

Lesson 32: Factorisation of polynomials (1): using the factor theorem....59
Finding a first factor by trial · dividing to obtain a quadratic factor · complete factorisation of cubics

Lesson 33: Factorisation of polynomials (2): comparing coefficients and special forms....60
Comparing coefficients to find the quadratic factor · sum and difference of two cubes

Lesson 34: Factorisation of polynomials (3): four or more terms....61
Grouping terms · quartics in disguise (quadratics in x^2) · repeated use of the factor theorem

Review Exercise: Polynomials....62
Mixed exam-style questions with answers covering the whole topic

Chapter 5: Rational Polynomial Functions (Partial Fractions)....63

Lesson 35: Partial fractions with distinct linear factors in the denominator....63
Proper fractions with distinct linear factors · the cover-up method · substitution method

Lesson 36: Partial fractions with a repeated factor in the denominator....65
Form $A/(x-a) + B/(x-a)^2 + C/(x-b)$ · finding constants by substitution and comparing coefficients

Lesson 37: Partial fractions with an irreducible quadratic factor....66
Form $(Bx + C)/(x^2 + k)$ for an irreducible quadratic factor · finding the constants

Lesson 38: Improper fractions where the degree of the numerator is greater than that of the denominator 67
Improper fractions · dividing first to obtain a polynomial plus a proper fraction · then splitting the remainder

Lesson 39: Improper fractions where the degrees are equal....68
Equal degrees · a constant quotient plus partial fractions

Lesson 40: Integration activities (mixed problems)....69
Mixed partial fraction problems · telescoping series using partial fractions

Review Exercise: Rational Polynomial Functions (Partial Fractions)....70
Mixed exam-style questions with answers covering the whole topic

Chapter 6: Mathematical Statements, Logic and Proofs....71

Lesson 41: Basic concepts in logic 71
Statements and non-statements · logical connectives and symbols · quantifiers ∀ and ∃

Lesson 42: Propositions 73
Simple and compound propositions · translating between words and symbols

Lesson 43: Truth value of a statement and its negation 74
Truth values T and F · negation of simple and quantified statements

Lesson 44: Composite statements: conjunctions 75
The conjunction $p \wedge q$ · its truth table · truth values of compound "and" statements

Lesson 45: Composite statements: disjunctions 76
The disjunction $p \vee q$ · inclusive "or" · De Morgan's laws · tautologies and contradictions

Lesson 46: Conditional statements, the converse, inverse and contrapositive 77
The conditional $p \Rightarrow q$ · converse, inverse and contrapositive · the biconditional $p \Leftrightarrow q$

Lesson 47: Direct proofs and proofs by counter-example 78
Direct proof using definitions · even and odd numbers · disproof by a single counter-example

Lesson 48: Proof by contradiction 79
Structure of proof by contradiction · irrationality of $\sqrt{2}$ · infinitely many / no largest arguments

Lesson 49: Proof by mathematical induction 80
Base case and inductive step · proving sum formulae · proving divisibility results

Integration activities (logic and proof) 81
Extra worked examples and model answers for Mathematical Statements, Logic and Proofs

Review Exercise: Mathematical Statements, Logic and Proofs 82
Mixed exam-style questions with answers covering the whole topic

Chapter 7: Binomial Expansions 83

Lesson 50: Generating Pascal's triangle 83
Building Pascal's triangle · reading coefficients · expanding $(a + b)^n$ for small n

Lesson 51: The general binomial theorem (positive integer powers) 85
Factorials and ${}^nC_r$ · the binomial theorem for positive integer n · expansions in ascending powers

Lesson 52: Finding specific terms and coefficients 86
The general term T_{r+1} · coefficient of a given power · term independent of x · finding unknown constants

Lesson 53: Binomial expansion for rational indices and convergence 87
$(1 + x)^n$ for negative and fractional n · range of validity $|x| < 1$ · expanding $(a + bx)^n$

Lesson 54: Binomial approximations 88
Approximating roots and powers with the first few terms · choosing x small · accuracy

Review Exercise: Binomial Expansions 89
Mixed exam-style questions with answers covering the whole topic

Chapter 8: Binary Relations 90

Lesson 55: Ordered pairs and the Cartesian product of two sets 90
Ordered pairs and their equality · the Cartesian product $A \times B$ · $n(A \times B) = n(A)n(B)$

Lesson 56: Binary relations and properties of relations (1) 91
Binary relations as subsets of $A \times A$ · listing relations · reflexive and symmetric relations

Lesson 57: Properties of relations (2) 92
Antisymmetric and transitive relations · testing all four properties

Lesson 58: Equivalence relations 93
Definition of an equivalence relation · proofs that relations are equivalences

Lesson 59: Equivalence classes 94
Equivalence class of an element · partition of a set into classes · congruence classes

Lesson 60: Order relations 95
Order relations · partial and total orders · comparing elements

Review Exercise: Binary Relations 96
Mixed exam-style questions with answers covering the whole topic

Chapter 9: Functions 97

Lesson 61: Definition of a function and notation 97
A function as a special relation · notation $f : A \to B$, $x \mapsto f(x)$ · images and pre-images

Lesson 62: Domain, codomain and range 99
Domain, codomain and range · natural domain of real functions · finding the range

Lesson 63: Types of mappings 100
One-to-one, many-to-one, one-to-many and many-to-many mappings · which are functions

Lesson 64: Injective and surjective mappings 101
Injective (one-to-one) functions · surjective (onto) functions · algebraic and graphical tests

Lesson 65: Bijective mappings 102
Bijective functions · restricting domains and codomains to make a bijection

Lesson 66: Inverse functions 103
Finding f^{-1} algebraically · domain and range of the inverse · graph as reflection in $y = x$

Lesson 67: Composite functions 104
Composite functions $f \circ g$ · order of composition · domain of a composite · inverse of a composite

Lesson 68: Algebra of functions 105
Sum, difference, product and quotient of functions · domains of combined functions

Lesson 69: Odd and even functions 106
Even functions $f(-x) = f(x)$ · odd functions $f(-x) = -f(x)$ · symmetry of graphs

Lesson 70: Monotonicity of functions 107
Increasing and decreasing functions · proofs from the definition · intervals of monotonicity

Lesson 71: Periodic functions 108
Periodic functions and their period · periods of sin kx, cos kx and tan kx

Lesson 72: Transformation of graphs: f(x + a) and f(x) + a 109
Horizontal translations $f(x + a)$ · vertical translations $f(x) + a$ · images of points

Lesson 73: Transformation of graphs: f(ax) and af(x) 110
Stretches $af(x)$ and $f(ax)$ · reflections $-f(x)$ and $f(-x)$ · combinations of transformations

Review Exercise: Functions 111
Mixed exam-style questions with answers covering the whole topic

Chapter 10: Indices, Surds and Logarithms 112

Lesson 74: Laws of indices and applications (1) 112
The laws of indices · zero, negative and fractional indices · evaluating and simplifying

Lesson 75: Laws of indices and applications (2): exponential equations 114
Exponential equations with a common base · equations reducible to quadratics

Lesson 76: Rational and irrational numbers 115
Rational and irrational numbers · converting recurring decimals to fractions · classifying numbers

Lesson 77: Simplification of surds 116
Laws of surds · simplest surd form · adding, subtracting and multiplying surds

Lesson 78: Rationalising the denominator 117
Rationalising $a/\sqrt{b}$ · using conjugates for $a + \sqrt{b}$ · square roots of surds

Lesson 79: Laws of logarithms (1) 118
Definition of a logarithm · converting between index and log form · special values

Lesson 80: Laws of logarithms (2) 119
Product, quotient and power laws · combining and expanding logarithms

Lesson 81: Common logarithms and change of base 120
Common logarithms · the change-of-base formula · $\log^a b = 1/\log_b a$

Lesson 82: Solving logarithmic and exponential equations (1) 121
Solving $a^x = b$ by taking logarithms · equations with different bases

Lesson 83: Solving logarithmic and exponential equations (2) 122
Logarithmic equations · checking for invalid solutions · equations with different bases

Lesson 84: Natural logarithms: rules of natural logarithms 123
e and natural logarithms · laws of ln · solving equations with e^x and ln x

Lesson 85: Graphs of $y = a^x$ and $y = \log_a x$ 124
Graphs of exponential and logarithmic functions · intercepts and asymptotes · inverse relationship

Lesson 86: Graphs of $y = e^x$ and $y = \ln x$ 125
Graphs of e^x and ln x · transformed graphs · domains and asymptotes

Lesson 87: Applications of the laws of logarithms 126
Exponential growth and decay models · compound interest · half-life · solving for time

Lesson 88: Definition of terms related to sequences and series 127
Sequence, term, general term u_n · finite and infinite sequences · recurrence relations · series

Review Exercise: Indices, Surds and Logarithms 128
Mixed exam-style questions with answers covering the whole topic

Chapter 11: Sequences and Series 129

Lesson 89: Arithmetic progression: the nth term and the arithmetic mean 129
Common difference · nth term of an AP · arithmetic mean · finding a and d

Lesson 90: Arithmetic progression: the sum of n terms (S_n) 131
Sum of n terms of an AP · using the last term · finding the number of terms · word problems

Lesson 91: Geometric progression: the nth term and the geometric mean 132
Common ratio · nth term of a GP · geometric mean · first term to exceed a value

Lesson 92: Geometric progression: the sum of n terms (S_n) 133
Sum of n terms of a GP · the two forms of the formula · growth problems

Lesson 93: Determination of u_n given S_n 134
$u_n = S_n - S_{n-1}$ · identifying the type of sequence from S_n

Lesson 94: Limit of a sequence or series 135
Limit of a sequence · convergent and divergent sequences · standard limits

Lesson 95: Convergence and divergence of a geometric series 136
Sum to infinity $a/(1 - r)$ · condition $|r| < 1$ · recurring decimals · series in x

Lesson 96: Sigma notation and summation of simple finite series 137
Sigma notation · rules of Σ · writing and evaluating finite series

Lesson 97: Summation of some standard series 138
Formulae for Σr, Σr^2, Σr^3 · series whose terms are polynomials in r

Lesson 98: Use of partial fractions in the summation of series (method of differences) 139
Method of differences · telescoping sums with partial fractions · sums to infinity

Review Exercise: Sequences and Series 140
Mixed exam-style questions with answers covering the whole topic

Chapter 12: Sets and Representation 141

Lesson 99: Equality of sets and comparability of sets 141
Equal sets · subsets and proper subsets · comparable and non-comparable sets

Lesson 100: Cardinality of a set and the power set 142
Cardinality n(A) · the empty set · the power set and 2^n subsets

Lesson 101: Universal set, set operations and De Morgan's laws (1) 143
Universal set · union, intersection, complement and difference · disjoint sets

Lesson 102: Set operations and De Morgan's laws (2) 144
De Morgan's laws · distributive laws · counting with $n(A \cup B)$

Lesson 103: Using Venn diagrams to prove relationships involving sets 145
Proving set identities with Venn diagrams · three-set problems · inclusion–exclusion

Lesson 104: Sets of numbers commonly used in mathematics (1) 146
$\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ · inclusions between number sets · classifying numbers

Lesson 105: Sets of numbers commonly used in mathematics (2): intervals 147
Interval notation (open and closed) · unions and intersections of intervals · closure of number sets

Review Exercise: Sets and Representation 148
Mixed exam-style questions with answers covering the whole topic

Chapter 13: Permutations and Combinations 149

Lesson 106: Fundamental counting principle and factorial notation 149
Multiplication and addition principles · factorial notation · simplifying factorials

Lesson 107: Definition and types of permutations 150
Arrangements of n objects (n!) · ^nP_r arrangements of r from n

Lesson 108: Permutations with repetition 151
Arrangements when repetition is allowed (n^r) · comparing with no repetition

Lesson 109: Permutations with some objects alike 152
Arrangements with identical objects $n!/(p!q!...)$ · words with repeated letters

Lesson 110: Circular arrangements 153
Circular arrangements (n − 1)! · necklaces (n − 1)!/2 · restrictions at round tables

Lesson 111: Selection of objects from a group (combinations) 154
Combinations ⁿCᵣ · selecting committees and teams · properties of ⁿCᵣ

Lesson 112: Permutations and combinations with restrictions 155
Objects together or apart · "at least" problems by complement · fixed positions

Review Exercise: Permutations and Combinations 156
Mixed exam-style questions with answers covering the whole topic

Chapter 14: Circular Functions (Trigonometry) 157

Lesson 113: Converting angles from degrees to radians and back 157
Radian measure · converting degrees ↔ radians · common angles in radians

Lesson 114: The six trigonometric ratios and ratios of special angles (1) 159
sin, cos, tan and their reciprocals · finding all ratios from one ratio

Lesson 115: Ratios of special angles (2) and the four quadrants 160
Exact values for 30°, 45°, 60° · signs in the four quadrants · related acute angles

Lesson 116: Perimeter and area of a circle; arc length 161
Circumference and area of a circle · arc length s = rθ · perimeter of a sector

Lesson 117: Area of a sector and area of a segment 162
Area of a sector ½r²θ · area of a triangle ½r² sin θ · area of a segment

Lesson 118: The sine rule 163
The sine rule · finding sides and angles · the ambiguous case

Lesson 119: The cosine rule 164
The cosine rule for sides and angles · area ½ab sin C · bearings problems

Lesson 120: The general angle 165
Angles of any size · related angle identities · negative angles and angles over 360°

Lesson 121: Pythagorean identities and proofs 166
sin²θ + cos²θ = 1 and its two variants · proving identities · simplifying expressions

Lesson 122: Graphs of sine, cosine and tangent (1) 167
Graphs of y = sin x, cos x, tan x · period, amplitude, zeros, symmetry and asymptotes

Lesson 123: Graphs of sine, cosine and tangent (2): y = a sin(bx + c) + d 168
y = a sin(bx + c) + d · amplitude, period, phase shift and centre line

Lesson 124: General solution of trigonometric equations (1): sin x = k 169
General solution of sin x = k · in degrees and radians · equations in sin(ax)

Lesson 125: General solution of trigonometric equations (2): cos x = k and tan x = k 170
General solution of cos x = k and tan x = k

Lesson 126: General solution of trigonometric equations (3): equations reducing to quadratics 171
Equations quadratic in sin, cos or tan · using identities to get one ratio

Lesson 127: Solving trigonometric equations in a given range 172
Listing solutions in a given interval · equations with multiple angles · checking all solutions

Lesson 128: Compound angle identities 173
sin, cos and tan of (A ± B) · exact values of 15°, 75°, 105° · using given ratios

Lesson 129: Double angle and half angle identities (1) 174
sin 2A, cos 2A, tan 2A · the three forms of cos 2A · power reduction · equations

Lesson 130: Double angle and half angle identities (2): the t-formulae 175
Half-angle results · the t-formulae t = tan(x/2) · proving identities and solving equations

Lesson 131: Factor formulae (1): sum to product 176
Sum-to-product formulae · factorising sums of sines and cosines · solving equations

Lesson 132: Factor formulae (2): product to sum 177
Product-to-sum formulae · proving identities with factor formulae

Lesson 133: Maximum and minimum values: a cos x + b sin x = R cos(x − α) 178
a cos x + b sin x = R cos(x − α) · maximum and minimum values · solving equations

Lesson 134: Small angles 179
Small-angle approximations for sin, cos, tan · using them in limits and estimates

Lesson 135: Inverse trigonometric functions (1): definitions and domains 180
Restricted domains · principal values of sin⁻¹, cos⁻¹, tan⁻¹ · exact values

Lesson 136: Inverse trigonometric functions (2): graphs and properties....181
Graphs of the inverse trigonometric functions · composite expressions · $\sin^{-1}x + \cos^{-1}x = \pi/2$

Lesson 137: Inverse trigonometric functions (3): equations and identities....182
Identities with inverse functions · solving equations involving inverse trigonometric functions

Review Exercise: Circular Functions (Trigonometry)....183
Mixed exam-style questions with answers covering the whole topic

Chapter 15: Coordinate Geometry....184

Lesson 138: Distance between two points and the midpoint of a line segment....184
Distance formula · midpoint formula · proving properties of triangles and quadrilaterals

Lesson 139: Coordinates of a point dividing a line internally and externally....185
Internal division in a ratio · external division · finding the ratio

Lesson 140: Gradient of a line and collinear points....186
Gradient $m = \Delta y/\Delta x = \tan\theta$ · angle with the x-axis · testing for collinearity

Lesson 141: Equation of a straight line....187
The forms of the equation of a line: $y = mx + c$, point–gradient, two-point, general and intercept forms

Lesson 142: Intersecting lines and the angle between two lines....188
Solving simultaneous linear equations for intersection points · angle between two lines

Lesson 143: Parallel and perpendicular lines....189
Conditions for parallel and perpendicular lines · perpendicular bisectors · feet of perpendiculars

Lesson 144: Distance of a point from a line....190
Perpendicular distance formula · distance between parallel lines · areas using distances

Lesson 145: Reduction to linear form (1)....191
Reducing non-linear laws to $Y = mX + c$ · choosing X and Y · finding constants from the line

Lesson 146: Reduction to linear form (2): using logarithms....192
Using logarithms for $y = ax^n$ and $y = ab^x$ · reading gradient and intercept

Lesson 147: Loci using Cartesian and parametric forms....193
Locus as a set of points · loci from distance conditions · Cartesian equations from parametric forms

Review Exercise: Coordinate Geometry....194
Mixed exam-style questions with answers covering the whole topic

Chapter 16: Matrices....195

Lesson 148: Definition of a matrix, order of a matrix and types of matrices....195
Matrix, element, order · row, column, square, zero, identity, diagonal and triangular matrices

Lesson 149: Algebra of matrices (1): equality, addition, subtraction and scalar multiplication....197
Equality of matrices · addition and subtraction · scalar multiplication

Lesson 150: Algebra of matrices (2): matrix multiplication....198
Conditions for multiplication · row-by-column rule · non-commutativity · powers of matrices

Lesson 151: The transpose of a matrix and determinants....199
Transpose and its properties · 2 × 2 and 3 × 3 determinants by expansion

Lesson 152: Properties of the determinant of a 3 × 3 matrix....200
Properties of determinants · row operations · zero determinants · det(kA) and det(AB)

Lesson 153: Cofactors and the adjoint of a 3 × 3 matrix....201
Minors and cofactors · the sign pattern · the adjoint as the transposed cofactor matrix

Lesson 154: Singular matrices and the inverse of a matrix....202
Singular and non-singular matrices · inverse of a 2 × 2 matrix · inverse of a 3 × 3 matrix

Lesson 155: Using matrices to solve systems of equations in 2 and 3 unknowns....203
Writing systems as $AX = B$ · solving with A^{-1} · 2 × 2 and 3 × 3 systems

Lesson 156: Applications of determinants and collinear points....204
Area of a triangle by determinant · collinearity test

Lesson 157: Matrix transformations: T(x, y) = (X, Y)....205
Linear transformations of the plane · images of (1, 0) and (0, 1) · standard matrices

Lesson 158: Matrix transformations: T(x, y, z) = (X, Y, Z)....206
Linear transformations in three dimensions · matrices from formulae · images of points

Lesson 159: Geometrical interpretation and solution structure of systems....207
Unique, no and infinitely many solutions · planes meeting at a point, a line, or not at all

Lesson 160: Invariant points and invariant lines....208
Invariant points · lines of invariant points · invariant lines through the origin

Lesson 161: Area and volume scale factors....209
|det M| as area and volume scale factor · orientation and negative determinants

Lesson 162: Combined transformations (1)....210
Order of combination (BA for "A then B") · describing a combined transformation

Lesson 163: Combined transformations (2) and inverse transformations....211
Inverse transformations · finding original points · combining rotations

Review Exercise: Matrices....212
Mixed exam-style questions with answers covering the whole topic

Chapter 17: Vectors....213

Lesson 164: Definition of terms and representation of vectors....213
Scalars and vectors · notation and representation · column and i, j, k forms · equal and negative vectors

Lesson 165: Magnitude and direction of a vector in 2D....215
Magnitude of a 2D vector · direction angle · resolving a vector into components

Lesson 166: Vectors in 3D: direction ratios and direction cosines....216
Magnitude in 3D · direction ratios · direction cosines and $l^2 + m^2 + n^2 = 1$

Lesson 167: Algebra of vectors....217
Addition, subtraction and scalar multiples · triangle and parallelogram laws · parallel vectors

Lesson 168: Unit vectors and position vectors....218
Unit vectors $\hat{a} = a/|a|$ · position vectors · $\overrightarrow{AB} = b - a$ · vectors of given length

Lesson 169: Properties of a line segment joining two points; the ratio theorem....219
The ratio theorem · midpoints · collinearity with vectors

Lesson 170: The scalar (dot) product and its properties....220
Definition $a \cdot b = |a||b| \cos\theta$ · component form · perpendicularity · angles in triangles

Lesson 171: The vector (cross) product and its properties....221
Vector product as a determinant · properties · areas of parallelograms and triangles

Lesson 172: Parallel and perpendicular vectors....222
Tests for parallel and perpendicular vectors · finding vectors perpendicular to two given vectors

Lesson 173: Vector equation of a straight line (given a point and a direction)....223
$r = a + \lambda d$ · checking whether a point lies on a line · points on a line

Lesson 174: Vector equation of a straight line through two fixed points....224
Line through two points $r = a + \lambda(b - a)$ · alternative forms

Lesson 175: Application of the scalar product: the angle between two lines....225
Angle between two lines from direction vectors · intersecting, parallel and skew lines

Lesson 176: Cartesian and parametric forms of the equation of a straight line....226
Parametric and Cartesian equations of a line in 3D · converting between forms

Lesson 177: Cartesian equation of a plane through three non-collinear points....227
Normal vector · $r \cdot n = a \cdot n$ · plane through three points using $\overrightarrow{AB} \times \overrightarrow{AC}$

Lesson 178: Parametric equation of a plane....228
$r = a + \lambda u + \mu v$ · converting parametric to Cartesian and back

Lesson 179: Distance of a plane from the origin and from any other point....229
Distance from the origin to a plane · distance from any point to a plane

Lesson 180: Lines and planes: the angle between a line and a plane....230
Angle between a line and a plane using the normal · sin θ formula

Lesson 181: Intersecting lines and planes....231
Point of intersection of a line and a plane · lines parallel to planes · lines lying in planes

Review Exercise: Vectors....232
Mixed exam-style questions with answers covering the whole topic

Chapter 18: Calculus 1: Limits and Differentiation....233

Lesson 182: Limits of functions and theorems on limits (1)....233
Meaning of lim f(x) · limits by substitution · indeterminate 0/0 forms by factorising · limit theorems

Lesson 183: Limits of functions (2): limits at infinity and with surds....235
Limits as $x \to \infty$ · dividing by the highest power · limits with surds using the conjugate

Lesson 184: Limits of trigonometric functions....236
lim sin x/x = 1 · related standard limits · trigonometric limits by manipulation

Lesson 185: Differentiation from first principles and standard functions (1)....237
The derivative as a limit · differentiating simple functions from first principles

Lesson 186: Differentiation of standard functions (2)....238
Power rule for any real n · derivatives of e^x, ln x, a^x, sin x, cos x, tan x · gradients at points

Lesson 187: Methods of differentiation (1): the chain rule....239
Function of a function · chain rule for powers, exponentials and logarithms

Lesson 188: Methods of differentiation (2): the product rule....240
The product rule · products with exponentials, logarithms and powers · simplifying derivatives

Lesson 189: Methods of differentiation (3): the quotient rule....241
The quotient rule · derivatives of rational functions · derivatives of tan, sec, cot

Lesson 190: Differentiation of trigonometric functions....242
Derivatives of sin, cos, tan of (ax + b) · powers of trigonometric functions · sec, cosec, cot

Lesson 191: Differentiation of inverse trigonometric functions....243
Derivatives of $\sin^{-1}x$, $\cos^{-1}x$, $\tan^{-1}x$ · chain rule with inverse trigonometric functions

Lesson 192: Second derivative and parametric differentiation (1)....244
The second derivative · verifying differential equations

Lesson 193: Differentiation of parametric equations (2)....245
dy/dx = (dy/dt)/(dx/dt) · second derivative for parametric curves · gradient at a point

Lesson 194: Implicit differentiation....246
Differentiating each term with respect to x · terms in y and xy · logarithmic differentiation

Lesson 195: L'Hôpital's rule....247
0/0 and ∞/∞ forms · applying L'Hôpital's rule once or repeatedly

Review Exercise: Calculus 1: Limits and Differentiation....248
Mixed exam-style questions with answers covering the whole topic

Chapter 19: Applications of Differentiation....249

Lesson 196: Rates of change, small changes and percentage change....249
Connected rates of change · small increments δy ≈ (dy/dx)δx · percentage errors

Lesson 197: Tangents and normals to a curve (1)....251
Equation of a tangent · equation of a normal · at a given point on a curve

Lesson 198: Tangents and normals (2)....252
Tangents with a given gradient · tangents from an external point · tangents to parametric curves

Lesson 199: Monotonicity of a function....253
Sign of f′(x) and monotonicity · finding increasing and decreasing intervals

Lesson 200: Determination of maximum and minimum turning points....254
Stationary points · first and second derivative tests · maximum/minimum problems

Lesson 201: Rolle's theorem and the mean value theorem....255
Rolle's theorem · the mean value theorem · finding the value c

Lesson 202: Curve sketching: domain of definition and intercepts....256
Domain of definition · intercepts with the axes · symmetry

Lesson 203: Curve sketching: stationary points and points of inflexion....257
Concavity · points of inflexion · stationary points of inflexion

Lesson 204: Curve sketching: asymptotes and the table of variations....258
Vertical, horizontal and oblique asymptotes · tables of variations

Lesson 205: Curve sketching: polynomial, rational and exponential functions (1)....259
Sketching polynomial curves · sketching rational curves using intercepts, asymptotes and stationary points

Lesson 206: Curve sketching (2): exponential and other functions....260
Sketching exponential and logarithmic curves · behaviour as x → ±∞

Review Exercise: Applications of Differentiation....261
Mixed exam-style questions with answers covering the whole topic

Chapter 20: Integration....262

Lesson 207: Integration as the reverse process of differentiation....262
Antiderivatives · the constant of integration · finding a curve from its gradient

Lesson 208: The indefinite integral: ∫xⁿ dx and ∫k dx 263
Integrating powers of x · rules for sums and multiples · simplifying before integrating

Lesson 209: Integration of axⁿ and (ax + b)ⁿ; definite integrals 264
∫(ax + b)ⁿ dx · definite integrals · area under a curve

Lesson 210: Integration of trigonometric functions 265
Integrals of sin, cos, sec² · using identities for sin²x, cos²x, tan²x · products to sums

Lesson 211: Integration of exponential and logarithmic functions 266
∫eˣ and ∫e^{ax+b} · ∫aˣ · ∫1/x and ∫1/(ax + b)

Lesson 212: Integration of fractions where the numerator is the derivative of the denominator 267
∫f′(x)/f(x) dx = ln|f(x)| · adjusting constants · ∫tan x and ∫cot x

Lesson 213: Integration of fractions using partial fractions 268
Splitting into partial fractions before integrating · definite integrals with logarithms

Lesson 214: Integration by substitution 269
Choosing a substitution · changing dx and the limits · common types

Lesson 215: Substitutions leading to 1/√(1 − x²) and 1/(1 + x²) 270
Standard integrals giving sin⁻¹ and tan⁻¹ · completing the square in the denominator

Lesson 216: Integration of modulus functions 271
Splitting the interval where the expression changes sign · areas with modulus functions

Lesson 217: Integration by parts (1) 272
The integration by parts formula · choosing u and dv/dx · products of polynomials with eˣ, sin, cos

Lesson 218: Integration by parts (2) 273
Parts with ln x · using parts twice · integrals that return to themselves

Review Exercise: Integration 274
Mixed exam-style questions with answers covering the whole topic

Formula Summary (Lower Sixth) 275

Algebra 275

Trigonometry 275

Coordinate geometry and vectors 275

Calculus 275

How to use these notes

These notes follow the official **MINESEC progression sheet** lesson by lesson. Each lesson has its own section, numbered exactly as in the progression, so you always know where you are in the syllabus.

Feature	What to do with it
Learning objectives (green box)	Read them first: they tell you what you must be able to do by the end of the lesson.
Lesson notes and Key Points (blue box)	Learn the definitions and formulas in the Key Points by heart.
Exam tips (yellow box)	Common mistakes and how GCE examiners expect answers to be set out.
Worked examples	Cover the solution, try the question yourself, then compare line by line.
Practice questions	Do them without looking at the answers. Mark your work and redo any you got wrong.
Review exercise	Mixed exam-style questions to test yourself on several lessons at once.

Need help? Ask the AsghinLabs AI Tutor at **www.asghinlabs.com/tutor.html** or message us on WhatsApp (+237 682 402 876).

Chapter 1: Theory of Quadratic Functions and Quadratic Equations

In this chapter you will learn to solve quadratic equations by three methods, use the relationship between the roots and the coefficients, and sketch and transform the graphs of quadratic functions.

Lesson	Title (MINESEC progression)
1	Quadratic expressions and equations
2	Method of completing the square
3	Quadratic formula method
4	Solving by factorisation
5	Relation between roots and coefficients of a quadratic equation
6	Symmetric properties of quadratic equations
7	Properties of quadratic functions
8	Graphs of quadratic functions
9	Simple transformation of the form $f(x) = ax^2$, $a \in \mathbb{R}$
10	Simple transformation of the form $y = f(x \pm a)$, $a \in \mathbb{R}$
11	Simple transformation of the form $y = f(x) \pm a$
12	Application of the nature of roots
13	Absolute value equations of the form $\lvert f(x) \rvert = a$

Lesson 1: Quadratic expressions and equations

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✓ define a quadratic expression and a quadratic equation
- ✓ write a quadratic equation in standard form and identify a, b and c
- ✓ check whether a given number is a root of a quadratic equation
- ✓ form a quadratic equation from a word problem

Lesson notes

A **quadratic expression** in x is an expression of the form $ax^2 + bx + c$, where a, b and c are real constants and **$a \neq 0$**. The highest power of x is 2.

When a quadratic expression is set equal to zero we get a **quadratic equation**: $ax^2 + bx + c = 0$.

The values of x that make the equation true are called its **roots** (or solutions). A quadratic equation has at most two real roots.

KEY POINTS

Standard form: $ax^2 + bx + c = 0$, with $a \neq 0$

a = coefficient of x^2; b = coefficient of x; c = constant term

- $x^2 - 5x + 6$ is quadratic ($a = 1$, $b = -5$, $c = 6$).
- $3x^2 - 12$ is quadratic ($a = 3$, $b = 0$, $c = -12$).
- $2x + 7$ is **not** quadratic (no x^2 term). $x^3 + x^2$ is **not** quadratic (the highest power is 3).

EXAM TIP

Always rearrange an equation into standard form (everything on one side, zero on the other) before you read off a, b and c. Keep the sign with each coefficient.

Worked examples

Example: Write $(x + 2)(2x - 3) = x - 1$ in standard form and state a, b and c.

Expand the left side: $2x^2 - 3x + 4x - 6 = 2x^2 + x - 6$

$2x^2 + x - 6 = x - 1$

$2x^2 + x - x - 6 + 1 = 0$

$2x^2 - 5 = 0$, so $a = 2$, $b = 0$, $c = -5$

Example: Show that $x = 3$ is a root of $2x^2 - 5x - 3 = 0$.

Substitute $x = 3$: $2(9) - 5(3) - 3 = 18 - 15 - 3 = 0$

The left side equals zero, so $x = 3$ is a root.

Example: The length of a rectangle is 3 cm more than its width. Its area is 40 cm². Form a quadratic equation for the width x cm.

Length = $(x + 3)$ cm, so area = $x(x + 3) = 40$

$x^2 + 3x - 40 = 0$ ($a = 1$, $b = 3$, $c = -40$)

(We solve this equation in Lesson 4: $x = 5$, so the rectangle is 5 cm by 8 cm.)

Practice Questions: Lesson 1

1. Which of these are quadratic expressions? (a) $4 - x^2$ (b) $5x - 2$ (c) $x(x + 1)(x - 1)$ (d) $(x + 3)^2 - x$
2. Write $(2x - 1)^2 = 3x + 5$ in standard form and state a, b and c.
3. Show that $x = -2$ is a root of $3x^2 + 5x - 2 = 0$.
4. $x = 1$ is a root of $2x^2 + kx - 5 = 0$. Find k.
5. The product of two consecutive positive integers is 72. Taking the smaller integer as n, form a quadratic equation in n.

Answers: *1. (a) and (d) only 2. $4x^2 - 7x - 4 = 0$; $a = 4$, $b = -7$, $c = -4$ 3. $3(4) + 5(-2) - 2 = 0$ 4. $k = 3$ 5. $n^2 + n - 72 = 0$*

Lesson 2: Method of completing the square

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ rewrite $x^2 + bx$ as a perfect square minus a constant
- ✔ solve a quadratic equation by completing the square, giving exact (surd) answers
- ✔ express $ax^2 + bx + c$ in the form $a(x + p)^2 + q$ and state its least or greatest value

Lesson notes

A **perfect square** is an expression such as $(x + 3)^2 = x^2 + 6x + 9$. Notice that the constant 9 is the square of half the coefficient of x. Completing the square means adding the right constant to make a perfect square.

KEY POINTS

$$x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2$$

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}$$

Steps to solve $ax^2 + bx + c = 0$ by completing the square:

- 1. Divide every term by a, so that the coefficient of x^2 is 1.
- 2. Move the constant term to the right-hand side.
- 3. Add (half the coefficient of x)^2 to **both** sides.
- 4. Write the left side as a perfect square.
- 5. Take the square root of both sides (remember ±), then solve for x.

Worked examples

Example: Solve $x^2 + 6x - 7 = 0$ by completing the square.

$x^2 + 6x = 7$

Half of 6 is 3; add $3^2 = 9$ to both sides: $x^2 + 6x + 9 = 16$

$(x + 3)^2 = 16$

$x + 3 = \pm 4$

$x = 1$ or $x = -7$

Example: Solve $2x^2 - 4x - 3 = 0$, leaving your answer in surd form.

Divide by 2: $x^2 - 2x - \frac{3}{2} = 0$

$x^2 - 2x = \frac{3}{2}$

Add 1 to both sides: $(x - 1)^2 = \frac{5}{2}$

$x - 1 = \pm\sqrt{\frac{5}{2}} = \pm\frac{\sqrt{10}}{2}$

$x = 1 \pm \frac{\sqrt{10}}{2}$

Example: Express $3x^2 + 12x + 5$ in the form $a(x + p)^2 + q$ and hence state the minimum value of the expression.

$3x^2 + 12x + 5 = 3(x^2 + 4x) + 5$

$= 3[(x + 2)^2 - 4] + 5$

$= 3(x + 2)^2 - 7$

Since $(x + 2)^2 \geq 0$, the minimum value is -7, when $x = -2$.

EXAM TIP

Completing the square is also used to find the turning point of a parabola (Lesson 7) and the centre of a circle (Upper Sixth).

Practice Questions: Lesson 2

1. Solve $x^2 - 8x + 5 = 0$ by completing the square, leaving your answers in surd form.
2. Solve $2x^2 + 6x - 1 = 0$ by completing the square.
3. Express $x^2 + 10x + 7$ in the form $(x + p)^2 + q$ and state its minimum value.
4. Express $5 - 6x - x^2$ in the form $a - (x + b)^2$ and state its maximum value and the value of x where it occurs.

Answers: *1. $x = 4 \pm \sqrt{11}$ 2. $x = (-3 \pm \sqrt{11})/2$ 3. $(x + 5)^2 - 18$; minimum -18 4. $14 - (x + 3)^2$; maximum 14 when $x = -3$*

Lesson 3: Quadratic formula method

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ derive the quadratic formula by completing the square
- ✔ solve any quadratic equation using the formula, giving exact or decimal answers
- ✔ recognise from $b^2 - 4ac$ when an equation has no real roots

Lesson notes

Completing the square on the general equation $ax^2 + bx + c = 0$ gives a formula that solves **every** quadratic equation.

Derivation:

$ax^2 + bx + c = 0$; divide by a: $x^2 + \frac{b}{a}x = -\frac{c}{a}$

Add $(\frac{b}{2a})^2$ to both sides: $(x + \frac{b}{2a})^2 = \frac{b^2}{4a^2} - \frac{c}{a} = \frac{b^2 - 4ac}{4a^2}$

$x + \frac{b}{2a} = \pm\frac{\sqrt{b^2 - 4ac}}{2a}$

KEY POINTS

Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

The quantity $b^2 - 4ac$ is called the **discriminant** (Δ). If $\Delta < 0$ the equation has no real roots.

EXAM TIP

Write down a, b and c with their signs before substituting. Most mistakes come from a wrong sign for b. Work out $b^2 - 4ac$ separately first.

Worked examples

Example: Solve $3x^2 - 4x - 2 = 0$, giving your answers to 3 significant figures.

$a = 3, b = -4, c = -2$

$b^2 - 4ac = 16 + 24 = 40$

$x = \frac{4 \pm \sqrt{40}}{6}$

$x = 1.72$ or $x = -0.387$

Example: Solve $x^2 + 2x - 1 = 0$, giving exact answers.

$b^2 - 4ac = 4 + 4 = 8$, and $\sqrt{8} = 2\sqrt{2}$

$x = \frac{-2 \pm 2\sqrt{2}}{2} = -1 \pm \sqrt{2}$

Example: Solve $5x^2 + 3x + 1 = 0$.

$b^2 - 4ac = 9 - 20 = -11 < 0$

The square root of a negative number is not real, so the equation has no real roots.

Practice Questions: Lesson 3

1. Solve $2x^2 - 3x - 4 = 0$, giving your answers to 3 significant figures.
2. Solve $x^2 - 6x + 4 = 0$, giving exact answers in simplest surd form.
3. Solve $4x^2 + 4x + 1 = 0$ using the formula. What do you notice?
4. A rectangle has width x m and length (x + 4) m. Its area is 30 m². Find x to 3 significant figures.

Answers: *1. 2.35, −0.851 2. $3 \pm \sqrt{5}$ 3. $b^2 - 4ac = 0$: one repeated root, $x = -½$ 4. $x = -2 + \sqrt{34} = 3.83$ (the negative root is rejected)*

Lesson 4: Solving by factorisation

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ use the rule "if PQ = 0 then P = 0 or Q = 0"
- ✔ factorise $ax^2 + bx + c$ by splitting the middle term and solve the equation
- ✔ use the difference of two squares and common factors to solve special cases

Lesson notes

If the product of two numbers is zero, at least one of them is zero: if PQ = 0, then P = 0 or Q = 0.

Steps:

- 1. Put the equation in the form $ax^2 + bx + c = 0$.
- 2. Find two numbers whose **product is ac** and whose **sum is b**.
- 3. Split the middle term using these two numbers.
- 4. Factorise in pairs.
- 5. Set each factor equal to zero and solve.

KEY POINTS

Special cases: $x^2 - a^2 = (x - a)(x + a)$ (difference of two squares); $ax^2 + bx = x(ax + b)$ (common factor)

EXAM TIP

Never divide both sides of an equation by x: you would lose the root x = 0. Factorise instead.

Worked examples

Example: Solve $x^2 - 5x + 6 = 0$.

Product 6, sum −5: the numbers are −2 and −3

$(x - 2)(x - 3) = 0$

$x = 2$ or $x = 3$

Example: Solve $2x^2 + 5x - 3 = 0$.

ac = −6; two numbers with product −6 and sum 5: 6 and −1

$2x^2 + 6x - x - 3 = 0$

$2x(x + 3) - 1(x + 3) = 0$

$(2x - 1)(x + 3) = 0$

$x = ½$ or $x = -3$

Example: Solve $6x^2 = 7x + 3$.

$6x^2 - 7x - 3 = 0$; ac = −18; the numbers are −9 and 2

$6x^2 - 9x + 2x - 3 = 0$

$3x(2x - 3) + 1(2x - 3) = 0$

$(3x + 1)(2x - 3) = 0$

$x = -⅓$ or $x = \frac{3}{2}$

Example: Solve $4x^2 - 9 = 0$ and $3x^2 - 12x = 0$.

$4x^2 - 9 = (2x - 3)(2x + 3) = 0$, so $x = \pm\frac{3}{2}$

$3x^2 - 12x = 3x(x - 4) = 0$, so $x = 0$ or $x = 4$

Example: Solve the rectangle problem from Lesson 1: $x^2 + 3x - 40 = 0$.

Product -40, sum 3: the numbers are 8 and -5

$(x + 8)(x - 5) = 0$, so $x = -8$ or $x = 5$

A width cannot be negative, so $x = 5$: the rectangle is 5 cm by 8 cm.

Practice Questions: Lesson 4

1. Solve $x^2 + x - 20 = 0$.
2. Solve $3x^2 - 11x + 6 = 0$.
3. Solve $9x^2 - 25 = 0$.
4. Solve $x(x - 4) = 21$.
5. Solve $5x^2 = 20x$.

Answers: *1. $x = 4$ or -5 2. $x = \frac{2}{3}$ or 3 3. $x = \pm 5/3$ 4. $x = 7$ or -3 5. $x = 0$ or 4*

Lesson 5: Relation between roots and coefficients of a quadratic equation

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ state and prove that $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$
- ✔ find the sum and product of the roots without solving the equation
- ✔ form a quadratic equation with given roots
- ✔ use the sum and product to find unknown coefficients

Lesson notes

Let α and β be the roots of $ax^2 + bx + c = 0$. Then

$$ax^2 + bx + c = a(x - \alpha)(x - \beta) = a[x^2 - (\alpha + \beta)x + \alpha\beta]$$

Dividing by a and comparing coefficients with $x^2 + (b/a)x + c/a$:

KEY POINTS

Sum of roots: $\alpha + \beta = -\dfrac{b}{a}$

Product of roots: $\alpha\beta = \dfrac{c}{a}$

An equation whose roots are α and β is $x^2 - (\alpha + \beta)x + \alpha\beta = 0$, i.e. $x^2 - (\text{sum})x + (\text{product}) = 0$

Worked examples

Example: Without solving, find the sum and product of the roots of $2x^2 - 6x + 1 = 0$.

$$\alpha + \beta = -\frac{-6}{2} = 3$$

$$\alpha\beta = \frac{1}{2}$$

Example: Form the quadratic equation with integer coefficients whose roots are $2/3$ and -1.

$$\text{Sum} = \frac{2}{3} - 1 = -\frac{1}{3};\ \text{product} = -\frac{2}{3}$$

$$x^2 + \frac{1}{3}x - \frac{2}{3} = 0$$

Multiply by 3: $3x^2 + x - 2 = 0$

Example: One root of $x^2 - kx + 12 = 0$ is three times the other. Find the possible values of k.

Let the roots be α and 3α

Product: $3\alpha^2 = 12$, so $\alpha^2 = 4$ and $\alpha = \pm 2$

Sum: $4\alpha = k$, so $k = 8$ or $k = -8$

Practice Questions: Lesson 5

1. Find the sum and product of the roots of $3x^2 + 7x - 2 = 0$.
2. Form the equation with integer coefficients whose roots are -2 and $5/2$.
3. One root of $x^2 + px + 12 = 0$ is 3. Find the other root and the value of p.
4. The roots of $x^2 - 6x + k = 0$ differ by 2. Find the roots and the value of k.

Answers: *1. sum −7/3; product −2/3 2. $2x^2 - x - 10 = 0$ 3. other root 4; p = −7 4. roots 2 and 4; k = 8*

Lesson 6: Symmetric properties of quadratic equations

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ recognise a symmetric expression in α and β
- ✔ write symmetric expressions in terms of α + β and αβ and evaluate them
- ✔ form equations whose roots are related to the roots of a given equation

Lesson notes

An expression in α and β is **symmetric** if swapping α and β does not change it, for example $\alpha^2 + \beta^2$ or $1/\alpha + 1/\beta$. Every symmetric expression can be written using only α + β and αβ, which we know from Lesson 5.

KEY POINTS

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$ $\quad \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$

$\alpha^2\beta + \alpha\beta^2 = \alpha\beta(\alpha + \beta)$

EXAM TIP

For an equation with "new" roots, find the new sum and the new product, then use $x^2 - (\text{sum})x + (\text{product}) = 0$.

Worked examples

Example: If α and β are the roots of $2x^2 - 3x - 4 = 0$, find $\alpha^2 + \beta^2$, $1/\alpha + 1/\beta$ and $\alpha^3 + \beta^3$.

$\alpha + \beta = \frac{3}{2}$, $\alpha\beta = -2$

$\alpha^2 + \beta^2 = \frac{9}{4} + 4 = \frac{25}{4}$

$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{3/2}{-2} = -\frac{3}{4}$

$\alpha^3 + \beta^3 = \frac{27}{8} - 3(-2)(\frac{3}{2}) = \frac{27}{8} + 9 = \frac{99}{8}$

Example: α and β are the roots of $x^2 - 4x + 1 = 0$. Find the equation whose roots are α^2 and β^2.

$\alpha + \beta = 4$, $\alpha\beta = 1$

New sum: $\alpha^2 + \beta^2 = 16 - 2 = 14$

New product: $\alpha^2\beta^2 = (\alpha\beta)^2 = 1$

Equation: $x^2 - 14x + 1 = 0$

Example: α and β are the roots of $x^2 + 3x - 5 = 0$. Find the equation whose roots are α + 2 and β + 2.

$\alpha + \beta = -3$, $\alpha\beta = -5$

New sum: $(\alpha + \beta) + 4 = 1$

New product: $\alpha\beta + 2(\alpha + \beta) + 4 = -5 - 6 + 4 = -7$

Equation: $x^2 - x - 7 = 0$

Practice Questions: Lesson 6

1. α and β are the roots of $x^2 - 5x + 3 = 0$. Find (a) $\alpha^2 + \beta^2$ (b) $(\alpha - \beta)^2$ (c) $\alpha/\beta + \beta/\alpha$ (d) $\alpha^3 + \beta^3$.
2. For the same equation, find the equation with integer coefficients whose roots are $1/\alpha$ and $1/\beta$.
3. α and β are the roots of $2x^2 + x - 3 = 0$. Find the equation whose roots are 2α and 2β.

Answers: *1. (a) 19 (b) 13 (c) 19/3 (d) 80 2. $3x^2 - 5x + 1 = 0$ 3. $x^2 + x - 6 = 0$*

Lesson 7: Properties of quadratic functions

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ state the shape of $y = ax^2 + bx + c$ from the sign of a
- ✔ find the vertex (turning point) and the axis of symmetry
- ✔ find the intercepts, the range, and where the function increases or decreases

Lesson notes

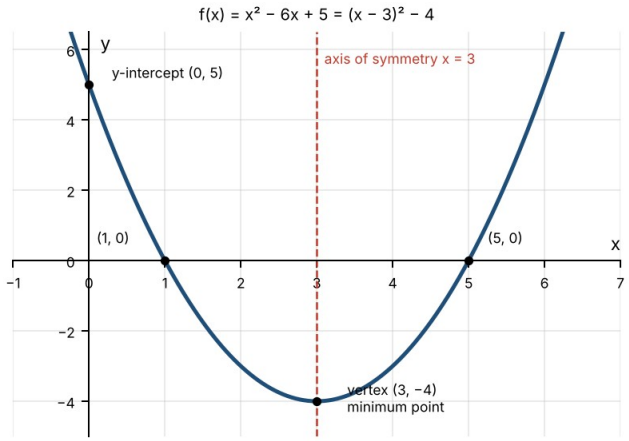
The function $f(x) = ax^2 + bx + c$ $(a \neq 0)$ is a **quadratic function**. Its graph is a curve called a **parabola**.

KEY POINTS

If $a > 0$ the parabola opens upwards (U) and has a **minimum** point. If $a < 0$ it opens downwards (∩) and has a **maximum** point.

Writing $f(x) = a(x - h)^2 + k$: the vertex (turning point) is (h, k) and the line of symmetry is $x = h = -\dfrac{b}{2a}$

- The **y-intercept** is $f(0) = c$.
- The **x-intercepts** are the real roots of $ax^2 + bx + c = 0$ (if any).
- The **domain** is all real numbers $\mathbb{R}$. The **range** is $f(x) \geq k$ if $a > 0$, or $f(x) \leq k$ if $a < 0$.
- **Symmetry:** points equally far either side of $x = h$ have the same y-value: $f(h + t) = f(h - t)$.
- **Increasing/decreasing:** if $a > 0$, f decreases for $x < h$ and increases for $x > h$ (the opposite if $a < 0$).



Worked examples

Example: For $f(x) = x^2 - 6x + 5$, find the vertex, the axis of symmetry, the intercepts and the range.

$f(x) = (x - 3)^2 - 9 + 5 = (x - 3)^2 - 4$

Vertex $(3, -4)$, a minimum point. Axis of symmetry: $x = 3$

y-intercept: $f(0) = 5$. x-intercepts: $(x - 1)(x - 5) = 0$, so $x = 1$ and $x = 5$

Range: $f(x) \geq -4$. f decreases for $x < 3$ and increases for $x > 3$

Example: Find the maximum value of $g(x) = 5 + 4x - 2x^2$ and the value of x where it occurs.

$g(x) = -2(x^2 - 2x) + 5 = -2[(x - 1)^2 - 1] + 5 = -2(x - 1)^2 + 7$

Maximum value 7 when $x = 1$

Practice Questions: Lesson 7

1. For $f(x) = x^2 + 4x - 5$, find the vertex, the axis of symmetry, the intercepts and the range.
2. Find the maximum value of $g(x) = -x^2 + 6x - 4$ and state the range of g.
3. The function $f(x) = x^2 + bx + c$ has its vertex at $(3, -2)$. Find b and c.
4. For which values of x is $f(x) = 2x^2 - 8x + 3$ increasing?

Answers: *1. vertex $(-2, -9)$; $x = -2$; y-intercept -5; x-intercepts 1 and -5; $f(x) \geq -9$ 2. maximum 5 (at $x = 3$); $g(x) \leq 5$ 3. $b = -6$, $c = 7$ 4. $x > 2$*

Lesson 8: Graphs of quadratic functions

LEARNING OBJECTIVES

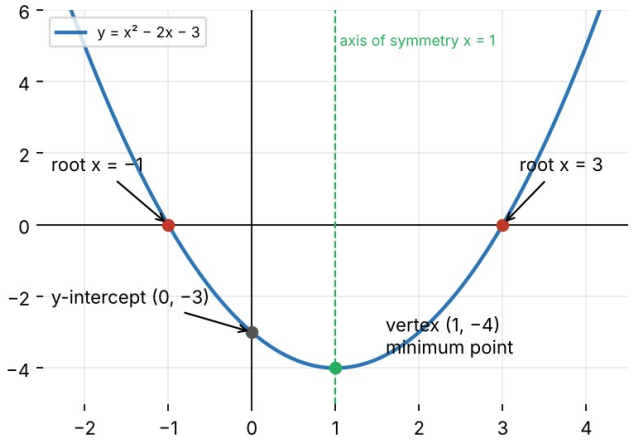
By the end of this lesson you should be able to:

- ✔ sketch the graph of a quadratic function showing its key points
- ✔ draw an accurate graph from a table of values
- ✔ use a graph to solve quadratic equations

Lesson notes

To sketch $y = ax^2 + bx + c$:

- 1. Decide the shape from the sign of a (∪ or ∩).
- 2. Find the y-intercept (0, c).
- 3. Find the x-intercepts by solving $ax^2 + bx + c = 0$ (if $\Delta < 0$ the graph does not cross the x-axis).
- 4. Find the vertex by completing the square or using $x = -b/(2a)$.
- 5. Plot these points and draw a smooth, symmetric curve. Label the key points.



KEY POINTS

The solutions of $ax^2 + bx + c = k$ are the x-coordinates of the points where the curve $y = ax^2 + bx + c$ meets the line $y = k$

Worked examples

Example: Sketch $y = -x^2 + 2x + 3$.

$a = -1 < 0$, so the shape is ∩

y-intercept: (0, 3)

x-intercepts: $-(x^2 - 2x - 3) = -(x - 3)(x + 1) = 0$, so (3, 0) and (−1, 0)

Vertex: $x = -\frac{b}{2a} = -\frac{2}{-2} = 1$; $y = -1 + 2 + 3 = 4$, so (1, 4) is the maximum point

Draw a ∩ curve through (−1, 0), (0, 3), (1, 4) and (3, 0).

Example: Draw a table of values for $y = x^2 - 2x - 3$ from $x = -2$ to $x = 4$ and use the graph to solve $x^2 - 2x - 3 = 0$.

x	−2	−1	0	1	2	3	4
y	5	0	−3	−4	−3	0	5

The curve meets the x-axis at $x = -1$ and $x = 3$, so these are the solutions (see the graph above).

Practice Questions: Lesson 8

1. Sketch $y = x^2 - 2x - 8$, showing the intercepts and the vertex.
2. Sketch $y = 4 - x^2$, showing the intercepts and the vertex.
3. Copy and complete the table for $y = 2x^2 - 3x - 2$, then draw the graph.

x	−2	−1	0	1	2	3
y	—	—	—	—	—	—

4. Use your graph from question 3 to solve (a) $2x^2 - 3x - 2 = 0$ (b) $2x^2 - 3x - 2 = 3$.

Answers: *1. ∪; x-intercepts −2 and 4; y-intercept −8; vertex (1, −9) 2. ∩; x-intercepts ±2; vertex and y-intercept (0, 4) 3. y = 12, 3, −2, −3, 0, 7 4. (a) x = −0.5 or 2 (b) draw y = 3: x = −1 or 2.5*

END OF FREE SAMPLE

Get the full notes

You have just read the opening of **Lower Sixth Mathematics**. The complete book has every lesson of the MINESEC progression sheet, in order:

- Every lesson, each with objectives, clear notes and KEY POINTS
- Worked examples and practice questions with answers
- Review exercises at the end of every chapter

This book: **3 000 FCFA** · Lower + Upper Sixth: **5 000 FCFA** per subject
Complete 5-subject A Level package: **20 000 FCFA**

Order at asghinlabs.com/academy

or WhatsApp **+237 682 402 876** · PDF sent straight to your phone